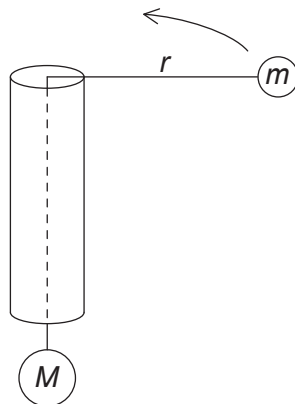


PAPER 1 / QUESTION 79

28 October 2020 / TZ0 / HL Paper 1 / original question 20 / 8820-6501

79. Mass M is attached to one end of a string. The string is passed through a hollow tube and mass m is attached to the other end. Friction between the tube and string is negligible.



Mass m travels at constant speed v in a horizontal circle of radius r . What is mass M ?

- A. $\frac{mv^2}{r}$
- B. mv^2rg
- C. $\frac{mgv^2}{r}$
- D. $\frac{mv^2}{gr}$